

Parity, Utility and Safety Are Not Independent

A Controlled Comparison of Three Bias-Mitigation Families
for Toxicity Classification

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<https://github.com/nigarrustamova/fairtox> · tag v1.0-final

The talk in four parts

- | | | |
|---|---|------------------|
| 1 | The problem, and the corpus | Rustamova, Nigar |
| 2 | How we measure, and what the errors are | Mirzayeva, Laman |
| 3 | The model, and the obvious fix | Samadov, Nijat |
| 4 | The mechanism, and what actually works | Aliyev, Orkhan |

The question

One corpus, one architecture, one recipe: how do mitigations at the three intervention points compare on parity — and what does each cost in utility and in safety?

38 runs · 115 models · 34.4 GPU-hours · every number here is generated from committed artefacts

Two errors — and they land on different people

False negative

Abuse gets through.

The person it targets is **unprotected**.

False positive

A benign comment is removed.

The person who wrote it is **silenced**.

Identity terms co-occur with abuse in scraped text — so the *token* becomes cheap evidence.

Then moderation is a tax on the communities it was deployed to protect.

Three places to intervene — rarely compared

PRE — the data

Rewrite the text so the term stops predicting the label.

Dixon 2018 · Park 2018

IN — the loss

Change the objective:
price the errors that matter.

Zhang 2018 · Sagawa 2020

POST — the threshold

Leave the model; move the boundary per group.

Hardt 2016

Each is normally evaluated **alone**, against its own baseline, on its own metric.

We run all three: five mitigations, three controls, eight seeds on the headline pair.

The corpus: Jigsaw Unintended Bias

Quantity	Value
Comments in the release	1,999,516
Identity-annotated (<i>our population</i>)	447,998 (22.4%)
Toxic rate	0.1134
Split: train / val / test	335,998 / 44,800 / 67,200
Subgroups configured	24
Clearing the FPR / FNR floor	14 / 9

- Target is the annotator *fraction* — binarised at 0.5, audit re-run at $\tau = 0.3 \dots 0.7$.
- Identity-annotated rows only: unrated $\neq$ identity-free. Licence CC0 1.0.

A baseline worth beating

Model	macro-F1	ROC-AUC	FPR	FNR
TF-IDF + logistic regression (balanced)	0.7460	0.9096	0.1179	0.2352
DistilBERT, fine-tuned	0.8003	0.9387	0.0289	0.4211

Balanced class weights on purpose — unweighted logistic regression reaches FNR 0.97.

Very different operating points: we quote both error rates, never macro-F1 alone.

Three axes — never combined into one score

Utility

macro-F1, ROC-AUC,
PR-AUC

is it a good classifier?

Parity

FPR gap, macro FPR,
EOD

who is wrongly censored?

Safety

subgroup FNR
whose abuse is missed?

$$\text{FPR}_k = \Pr(\hat{p} \geq \tau \mid y = 0, k) \quad \text{gap} = \max_k \text{FPR}_k - \min_k \text{FPR}_k$$

A single “is it better?” number would hide the trade-off we exist to measure.

Five guards, all fixed before we saw a result

- **Separate support floors** ($N \geq 100$ each) — **14 subgroups for parity, 9 for safety.**
- **Wilson intervals** — a bootstrap returns $[0, 0]$ at zero false positives, whatever the sample size.
- **Collapse check** — a model that flags nothing has **perfect parity**. None of ours collapsed.
- **One split per seed**, fingerprinted; duplicates measured: $\leq 1.11\%$ of test rows, effect ≤ 0.0032 .
- **Outcomes A / B / C / D pre-registered** in a config file before any model was trained.

What the model actually gets wrong

What the comment actually is	<i>n</i>	%	Label arguable
hostile, but not aimed at the group	21	35.0	11
genuinely derogatory towards the group	16	26.7	16
argues <i>against</i> bigotry	12	20.0	0
neutral mention, no hostility	7	11.7	0
quotes hostility in order to criticise it	4	6.7	0

a uniform random sample of 60 identity-bearing false positives, read and coded by hand

45% arguable

a reasonable annotator could have called them toxic

38% protected speech

counter-speech, quotation, neutral mention

Two of the sixty

“He was mentally-ill.”

model score 0.984 · dataset label: non-toxic · **neutral mention**

“Yes, they condemn Islamic terrorists, who are no different than Nazis and White Supremacists . . .”

model score 0.789 · dataset label: non-toxic · **counter-speech**

The worst error a moderation system can make: silencing the speech the policy exists to protect.

The model

Architecture

DistilBERT, pretrained

[CLS] $\rightarrow$ dropout $\rightarrow$ Linear(768, 1)

DistilRoBERTa for the backbone arm — no code change

Recipe

AdamW · lr 3×10^{-5} · batch 64 · 3 epochs
max length 128 · mixed precision

best-epoch selection, early stopping — identical in both arms

No `AutoModelForSequenceClassification`: it hides the loss, and every intervention here lives in the loss.

115 models — and how much of a difference is noise

The budget we used

One MIG 3g.20gb slice of an A100
0.064–0.071 s/step · peak 3.99 GB · 34.4 GPU-h

spare memory went to more seeds, not a bigger batch

The noise floor

The same baseline trained **five** times
lands in **two** states, not five

46 of 67,200 test decisions differ — 0.068%

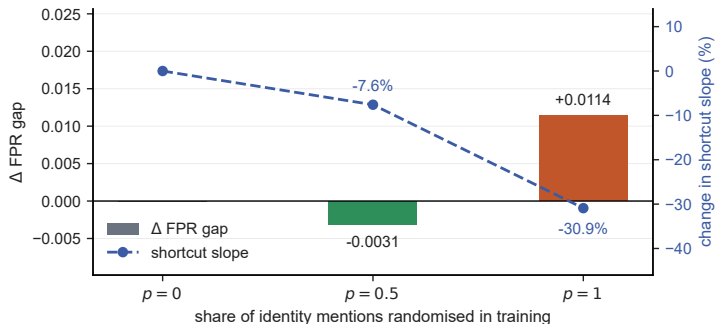
Floor 0.0000 on the FPR gap and 0.0017 on the worst-moving metric — about 78× smaller than the effect we report.

The obvious fix: swap the identity term

*“I am a proud **Muslim** woman”* $\longrightarrow$ *“I am a proud **Christian** woman”*

- Swapped **in place**, never appended — otherwise the step count changes too.
- Two doses: $p = 0.5$ and $p = 1.0$, five seeds each.
- Assumes toxicity is invariant under the swap — true for abuse, false for facts.

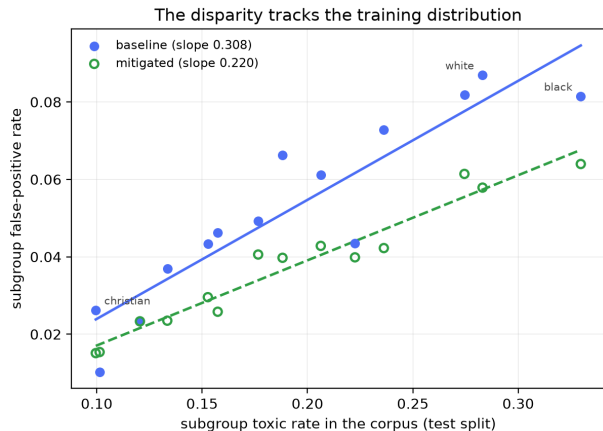
It does not work — and the way it fails is the result



five seeds per dose; the slope is the shortcut regression of the next section

At full dose the shortcut flattens 30.9% and the gap still does not close: the token was never the whole cause.

So where does the gap come from?



A subgroup's **toxic rate** predicts the model's **FPR** on that subgroup almost perfectly.

$r = 0.885\text{--}0.925$ in *every* configuration we ran.

Not prejudice — a conditional probability learned from the corpus and applied as a prior.

The fix: one *asymmetric* weight

	non-toxic	toxic
background	1	1
identity-bearing	α	1

$$\mathcal{L} = \frac{\sum_i w_i \ell_i}{\sum_i w_i} \quad \alpha = 4$$

- One TRAIN function; the arms differ in **one argument**.
- Split and training fingerprints must match — otherwise the comparison is **refused**.
- α fixed in advance, swept in the ablation.

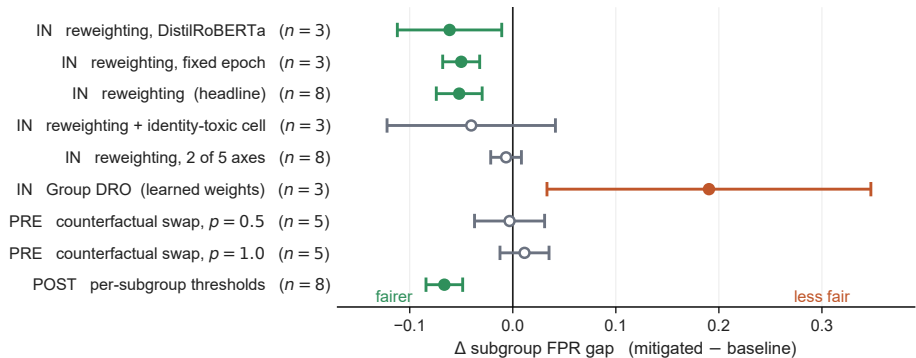
The headline, over eight seeds

Axis	Metric	Baseline	Mitigated	Δ	95% CI	Seeds
Parity	subgroup FPR gap	0.1072	0.0552	-0.0520	[-0.0742, -0.0297]	8/8
Utility	macro-F1	0.8031	0.7903	-0.0128	[-0.0153, -0.0104]	8/8
Safety	subgroup FNR	0.4382	0.5520	+0.1138	[+0.0824, +0.1452]	8/8

- 8/8 seeds agree in sign · sign test $p = 0.0078$ · pre-registered outcome **C**.
- The shortcut slope flattens 44.6% — and the fit stays just as tight.
- Threshold-free metrics agree: BPSN +0.0114, BNSP -0.0169.

The price: 2.19 points of missed abuse per point of FPR gap closed.

The three intervention points differ in *sign*



filled marker = the paired 95% interval excludes zero; three-seed families carry a direction only

Three controls, one conclusion

Control: what it weights	Seeds	FPR gap		subgroup FNR	
		before	after	before	after
inverse class frequency (<code>c1ass</code>)	3	0.1152	0.3005	0.4183	0.2039
identity-toxic cell alone	1	0.0767	0.1374	0.4744	0.3374
Group DRO, weights <i>learned</i>	3	0.1153	0.3056	0.4186	0.1877

Training cell	Ours	Learned
background, non-toxic	0.143	0.000
background, toxic	0.143	0.150
identity, non-toxic	0.571	0.440
identity, toxic	0.143	0.410

The claim

The gain needs an **asymmetric** weight on the identity *non-toxic* cell alone. Raise the *toxic* cell too — by hand or learned — and the frontier runs backwards.

What we cannot claim

- Absolute parity improved; `white/christian` relative parity did **not** ($3.32\times \rightarrow 3.83\times$).
- One subgroup got worse: `atheist` +50% from a small base.
- 45% of the coded false positives are **annotator disagreement** — $n = 60$, one coder.
- Support is uneven: 14 of 24 subgroups for parity, 9 for safety.
- Only two of eight configurations reach eight seeds.
- One corpus, one language, coarse binary identity categories.

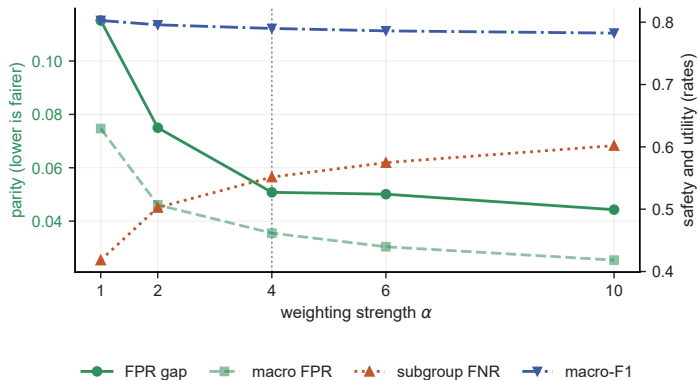
Conclusion

- The disparity is a **property of the training distribution**, not a bug in one model.
- It can be reduced — -0.0520 over eight seeds — but never for free: 2.19 points of missed abuse per point.
- The three families differ in **sign**, not merely in strength.
- Three controls say why: the gain needs an *asymmetric* weight, so even a learned worst-group objective cannot find it.

Thank you — questions?

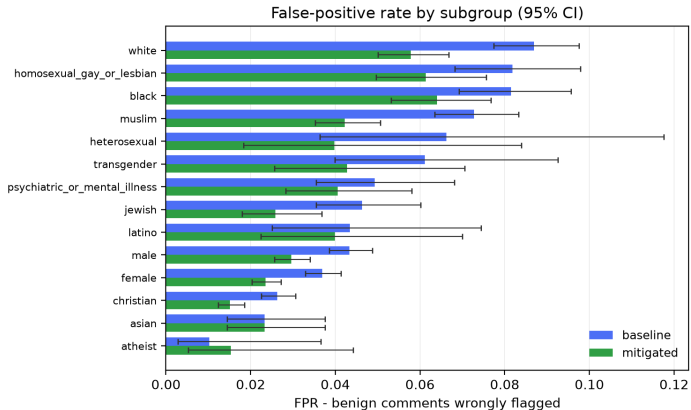
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How strong should the weighting be?



three seeds; parity flattens after the pre-registered $\alpha = 4$ while the safety cost keeps climbing

Who the intervention moves



12 of the 14 improve; *asian* is unchanged and *atheist* — the smallest group — gets worse